Chapter 01 – Interaction Design

Summary

1. Introduction to Interaction Design (IxD)

Definition: Interaction Design (IxD) focuses on designing user experiences that enhance how people interact with technology in their daily lives—whether at work, in communication, or in play.

It extends beyond traditional human-computer interaction (HCI) by considering emotional, practical, and contextual aspects of use.

- **Emotional aspect** - how the technology makes users feel (e.g., frustration, joy, trust)
- **Practical aspect** - how well it supports their goals or tasks
- **Contextual aspect** - the environment or situation in which the technology is used (e.g., mobile use on the go vs. desktop in an office)

- **Key Quotes**:
  - Sharp, Rogers, and Preece (2019): "Designing interactive products to support the way people communicate and interact in their everyday and working lives."
  - Winograd (1997): "The design of spaces for human communication and interaction."

2. Learning Objectives

- Differentiate between good and poor interaction design.
- Evaluate the pros and cons of digitizing activities (e.g., ticket machines, parking apps).
- Understand the relationship between usability and user experience (UX).
- Explore accessibility and inclusiveness in design.
- Learn about multidisciplinary collaboration in IxD.

3. Core Concepts

- **People-Centered Design** (Don Norman):

    - Focuses on specific user needs and capabilities.
    - Principles:
        1. Solve the right problem (address root causes).
        2. Prioritize people.
        3. Adopt a systems view (consider interdependencies).
        4. Continuously test and refine designs.
- **Good vs. Bad Design**:
    - Example: TiVo remote (ergonomic, intuitive) vs. poorly designed elevator controls (lack of visibility).

4. Digital Transformation

- **Benefits**: Convenience, speed, and efficiency (e.g., parking apps, QR codes).
- **Drawbacks**: Excludes non-smartphone users, privacy concerns, and loss of human interaction.

5. Key Design Considerations

- **What to Design**:
    - Understand users, activities, and contexts.
    - Optimize interactions to match user needs (e.g., IoT devices like smart doorbells).
- **Interdisciplinary Nature**:
    - Combines psychology, social sciences, engineering, and design practices (graphic, industrial, etc.).
    - Fields like HCI, ubiquitous computing, and cognitive ergonomics contribute to IxD.

6. User Experience (UX) and Usability

- **UX Definition**: How users perceive and emotionally respond to a product (e.g., smartwatch aesthetics).
    - **Hassenzahl's Model**:
        - **Pragmatic**: Functionality and ease of use.
        - **Hedonic**: Emotional and experiential aspects (e.g., joy,

creativity).

- **Usability Goals**:
  - Effectiveness, efficiency, safety, utility, learnability, memorability, and satisfaction.
- **UX Goals**:
  - **Desirable**: Fun, engaging, rewarding.
  - **Undesirable**: Frustrating, deceptive, boring.

7. Accessibility and Inclusiveness

- **Accessibility**: Ensures products are usable by people with disabilities (e.g., screen readers).
- **Inclusiveness**: Designs for diverse users (age, income, disability).
- **Types of Disabilities**:
  - Sensory (vision/hearing loss), physical (mobility), cognitive (memory).
  - Permanent, temporary, or situational (e.g., noisy environments).

8. Design Principles

- **Visibility**: Make actions obvious (e.g., elevator card slots).
- **Feedback**: Provide responses to user actions (e.g., button highlights).
- **Constraints**: Prevent errors (e.g., key shapes).
- **Consistency**: Uniformity in interfaces (e.g., Ctrl+S for save).
- **Affordances**: Perceived clues for use (e.g., scrollbars, buttons).

9. Challenges and Trade-offs

- **Cultural Differences**: Date formats, color symbolism, and interface preferences.
- **Consistency Issues**: Conflicting shortcuts (e.g., Ctrl+S for save vs. spell-check).
- **External Inconsistency**: Varied keypad layouts (phones vs. calculators).

10. Key Takeaways

- IxD is multidisciplinary, focusing on user needs, contexts, and experiences.
- Balancing usability (functionality) and UX (emotion) is critical.
- Principles like feedback and affordances guide effective design.
- Inclusive design ensures accessibility for diverse populations.